

Modeling Measures for Evaluating LLM Explanation Faithfulness

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Abstract

As artificial intelligence systems become more advanced, there is a growing need to explain those systems. However, there is no consensus on how to ensure that those explanations are accurate to the system’s internal processes. Many different measures have been proposed that each make different assumptions, evaluate specific types of explanations, and require different levels of model access. This can make it difficult to determine which measure is appropriate for a given situation. To address this challenge, we introduce the Explanation Faithfulness Evaluation Measures Ontology (EFEMO). This ontology provides a schema to compile, compare, and recommend faithfulness evaluation measures based on the proxy characteristics used, assumptions made, and other properties. It was built using the Ontology Engineering methodology and is publicly available¹. We evaluate EFEMO using a set of competency questions based on the use case of an explainability researcher.

Keywords

Ontology, Explanation Evaluation, Faithfulness

1. Introduction

The use of artificial intelligence (AI) has rapidly increased with the rise of LLM-based chatbots such as ChatGPT [1]. This has raised questions about where to use AI systems and how much to trust them. Explainable artificial intelligence (XAI) tries to answer those questions by developing AI systems that either explain themselves or explain other AI systems. These try to improve users trust in the system [2, 3, 4], allow non-AI experts to make informed choices when using the system [5, 6], and fulfill legal requirements on the system’s deployment [7, 8, 9]. This is especially important in fields such as healthcare and finance, where decisions can have a significant impact on peoples lives.

However there is no consensus on what constitutes an explanation. We use the definition presented in [10]: an *explanation* is an account of an AI system, including the implicit and explicit knowledge used in its reasoning processes and the specific decision. These may be in the form of natural language, heat maps of input, or analysis of internal values. An important property of explanations is their *faithfulness*, whether or not they accurately represent the reasoning process behind a model’s prediction [11]. Faithful explanations encourage users to trust the system without worrying about hidden biases or spurious correlations affecting decisions.

Despite its importance and the high number of papers published on the topic, there is currently no way to calculate a ground truth faithfulness score for a given explanation [11]. Instead, measures must make assumptions about how a model works and measure proxy characteristics that are related to faithfulness. It is usually easier for these measures to show an explanation is unfaithful than to prove it is faithful [12, 13]. This has led to a large number of measures being developed. These vary in many areas: the scope of the evaluation [14], the amount of model access required [15], the type of explanation required [16], the proxy characteristic measures [17], and whether the measure gives a binary or graded score [11]. It is also unclear how faithful an explanation needs to be for a given scenario, and how to evaluate whether it has met that level [12, 18]. The quality of these measures has also been brought into question, especially when humans score poorly on them [19]. This makes it

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even more difficult for researchers to determine which method to use and how to compare them.

We introduce the Explanation Faithfulness Evaluation Measures Ontology (EFEMO) to help solve this problem. EFEMO provides a common structure to organize and compare these evaluation measures. It also serves as the schema for a knowledge graph of faithfulness evaluation measures that can be used to recommend specific measures to researchers. EFEMO was developed following the Ontology Engineering method [20] and is evaluated using a use case and 10 competency questions.

2. Related Work

A significant issue in XAI is that the generally accepted definition of faithfulness is not precise. It does not provide any indication of how to evaluate faithfulness or what implementations should be used [21, 22]. This has led to the question of whether proving faithfulness is possible. One argument is that faithfulness can only be proven using mechanistic interpretability techniques and any other techniques introduce abstractions that prevent accurate determinations [23]. Others argue that faithfulness can be proven, so long as certain assumptions about how AI models and explanations hold [11]. The validity of various assumptions is still under debate [23, 24, 25].

The variety in explanation types and the debate over the possibility of proving faithfulness has led to a huge number of faithfulness evaluation measures. The ones proposed by [26] evaluate feature attribution explanations and assume that models make similar decisions only if their reasoning is similar. [27] makes similar assumptions but evaluates natural language assumptions and requires some access to the model's internals. Other methods, such as [28], are dataset-specific. This large variety of measures makes it difficult to determine which method is appropriate for a given scenario.

Other researchers have also considered the problem of multiple explanation faithfulness evaluations. [23] considers the claims that faithfulness researchers often make and argues that they should not be assumed. Similarly, [29] argues that the assumptions commonly made lead to measures evaluating explanation self-consistency rather than faithfulness. It has also been shown that faithfulness measures that initially seem to measure the same thing produce very different scores for the same explanations [30]. [31] evaluates a few faithfulness measures specifically based on their diagnosticity, i.e. whether the measure can identify obviously synthetic or unfaithful explanations [30].

We also considered explainability work that is not specific to faithfulness. Surveys such as [15, 32, 14, 33] look at the entirety of explanation evaluation, which help provide information on what the field finds important in an evaluation measure. Other surveys are even more general and look at explainability techniques [17, 34]. These are particularly useful for considering 'inherently interpretable' methods whose explanations were not evaluated. [34, 13, 35] provide simple taxonomies for explainability methods and evaluation methods that can be reused for future modeling.

It should be noted that work has been on modeling explainability to promote its use. The explanation ontology [36] helps system designers determine what explanations to include. It includes modeling of explanations and users, which can be reused for modeling explanation evaluation measures.

3. The Explanation Faithfulness Evaluation Measures Ontology

3.1. Development Methodology

We began with a literature analysis to determine what explanation faithfulness measures have been proposed and the qualities in an evaluation method that the XAI community considers important. The process began by searching for papers that cite [11] using Google Scholar, or are cited by that paper. For each paper found using this method we look for prominent citations and newer works citing the paper. Continuing this chaining should give us a broad set of papers; we can then exclude those papers that do not meet our criteria for relevancy and quality. Since our search process is centered around [11], we expect most papers we gathered to use their definition of faithfulness. We did read the abstract and introduction of each paper to confirm this, and exclude those that do not. To focus our work, we

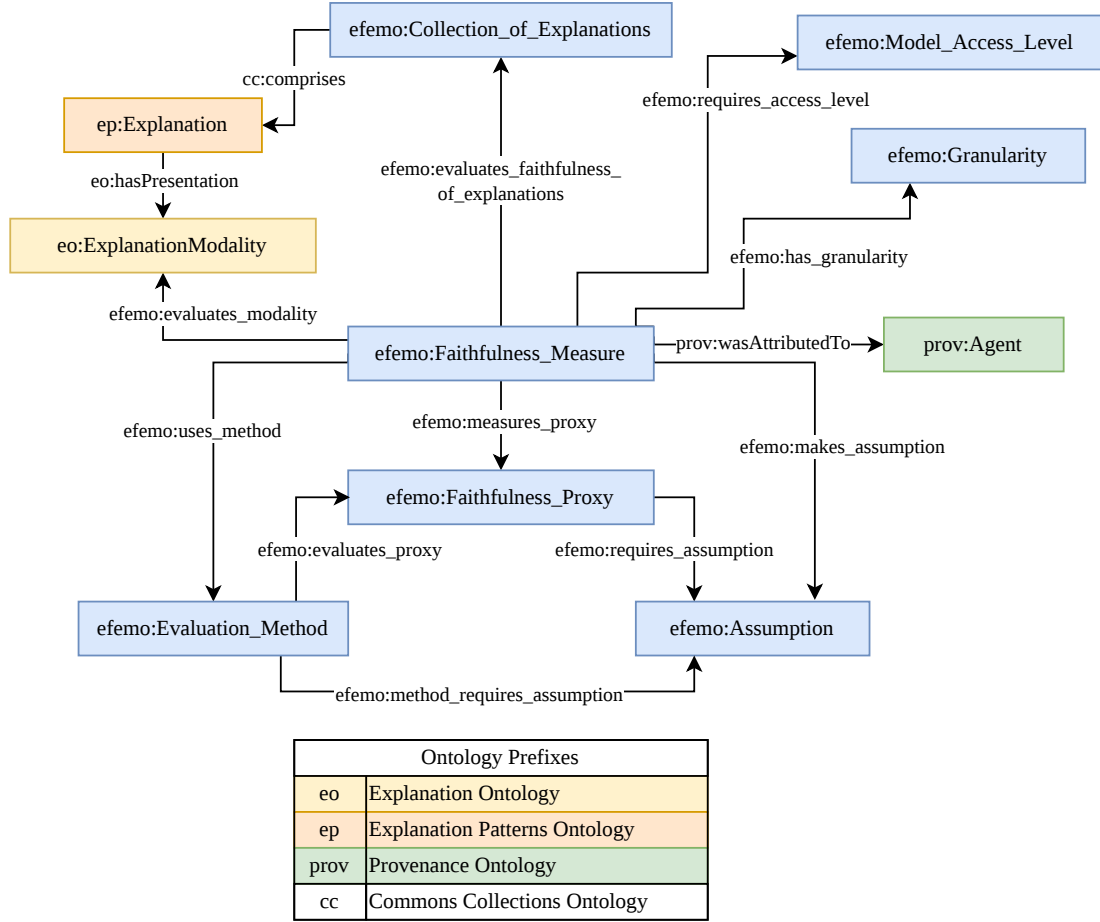


Figure 1: An overview concept diagram of EFEMO.

only included papers that evaluated text-based explanations or used text-based tasks. This literature analysis identified 46 surveys and 101 measure-proposing papers.

Once we had a comprehensive view of explanation faithfulness in the literature, we began to develop an ontology of measure properties and how they compare. Following the process described in [20], we created the use case of a researcher searching for an appropriate faithfulness evaluation measure. This first version of EFEMO is focused on the properties of explanation faithfulness evaluation measures found in the literature in order to facilitate comparison and situation applicability. These were pulled from various surveys discussing faithfulness such as [17, 15, 34, 13].

3.2. Ontology Design

A high-level concept diagram of our ontology is shown in Figure 1. To promote interoperability we reuse concepts from the Explanation Ontology [36], the Explanation Patterns Ontology [37], the Provenance Ontology [38], and the Commons Collections Ontology [39]. Faithfulness measures evaluate sets of explanations (though a set may only have a single explanation). Those explanations have a specific format, or modality, that the faithfulness measure expects. Since there is currently no gold-standard method of determining faithfulness, every measure uses a proxy characteristic. These may be self-consistency [29], robustness equivalence [13], or sufficiency [26]. We also record which assumptions different faithfulness measures make, and use superproperty chains to allow inferring which assumptions must hold based on the proxy characteristic and evaluation method used [11, 23]. We also track the

Table 1
Competency questions used to validate EFEMO

Competency Questions	
Focus:	Evaluation Measures
CQ 1:	What article is measure M attributed to?
CQ 2:	What explanation modality, granularity, and evaluation method does measure M use?
CQ 3:	What faithfulness assumptions are required to measure proxy P?
Focus:	Measure Recommendations
CQ 4:	What measures evaluate explanation modality D at granularity G using proxy characteristic P?
CQ 5:	What measures require model access level L while using an evaluation method of type E?
CQ 6:	Are there any measures with the same explanation modality, granularity, and model access requirements as measure M?
Focus:	Analysis of Measures
CQ 7:	Do measures M and N share any assumptions?
CQ 8:	What assumption is most commonly made when evaluating explanations with modality D?
CQ 9:	List the proxy characteristics used in evaluations that require model access level L by popularity.
CQ 10:	What explanation modality is evaluated by the fewest measures (of those included in the system)?

provenance of every measure using the PROV ontology [38]. The granularity of the measure is based on how many explanations are needed to use the measure, and the associated access level represents how much (if any) internal model access is necessary to evaluate that explanations from that model.

3.3. Ontology Evaluation

To ensure our ontology can meet the needs of an XAI researcher, we created a set of competency questions [40]. These questions, shown in Table 1, cover the core aspects of evaluation measures, measure recommendations, and measure analysis. The evaluation measure questions were chosen to ensure that there is enough information on each faithfulness measure for it to be a useful reference for researchers. Properties such as explanation modality, granularity, and required assumptions are given in articles proposing new measure because that information is critical to determining when a measure can be reused. The measure recommendation questions were chosen The analysis questions were chosen because these and similar questions often appeared in survey papers comparing explanation faithfulness measures [30, 31] or analyzing the field overall [11, 19, 14]. A knowledge graph based on EFEMO should be able to answer these questions when formulated as SPARQL queries¹

The ontology was designed so that additional characteristics of faithfulness measures could be easily added in order to answer future competency questions. The current characteristics require between 1 and 6 object properties (and appropriate inverse properties) to describe their relationships with other characteristics, with an average of 2.6 properties. We anticipate additional characteristics will require a similar number of properties.

We used Protégé to check the consistency of the ontology, and we used the OOPS! Ontology Pitfall Scanner [41] to evaluate the quality of the ontology. The following pitfalls were found:

- 5 unconnected ontology elements: 4 from Commons ontologies [39], and 1 from the Explanation Ontology [36]
- 33 missing annotations: 4 from the Explanation Ontology [36], 6 from the Explanation Patterns Ontology [37], and 23 from the Provenance Ontology [38]
- 41 missing domains or ranges in properties: 34 from Commons ontologies [39], 2 from the Provenance Ontology [38], 2 from the Explanation Ontology [36], and 3 from the Explanation Patterns Ontology [37]

¹Example SPARQL queries are shown on our website: https://tetherless-world.github.io/faithfulness-measures-ontology/cq_queries

- 52 missing inverse relationships: 42 from the Provenance Ontology [38], 5 from Commons ontologies [39], 2 from the Explanation Ontology [36], and 3 from the Explanation Patterns Ontology [37]
- 2 properties with multiple domains or ranges, both from the Provenance Ontology [38]
- Different naming conventions used in the ontology
- 2 sets of equivalent classes not explicitly declared

The vast majority of pitfalls that OOPS! detected were inherited from the ontologies we reused. The different naming conventions pitfall is because the reused ontologies use different naming conventions from EFEMO (ex. the Provenance Ontology uses camel case, while EFEMO uses snake case). It is unclear why the final pitfall, equivalent classes, was flagged since none of the classes identified are used in EFEMO and only appear in the reused ontologies. Based on these results of the OOPS! validation tool, we consider our ontology to be free of common pitfalls.

4. Conclusions and Future Work

The current version of the Explanation Faithfulness Evaluation Measures Ontology can support researchers in finding an appropriate faithfulness measure to use in their work. It was created using the Ontology Engineering methodology based on a literature analysis of faithfulness evaluation methods. It has been evaluated using competency questions and OOPS! and is publicly available.

In the future we plan on developing a tool for creating a knowledge graph based on EFEMO by extracting evaluation measures from articles. Since we currently only support evaluations for text-based tasks, we could also expand EFEMO to include measures evaluating the faithfulness of image-based and tabular-based explanations.

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Declaration on Generative AI

The authors have not employed any Generative AI tools.

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